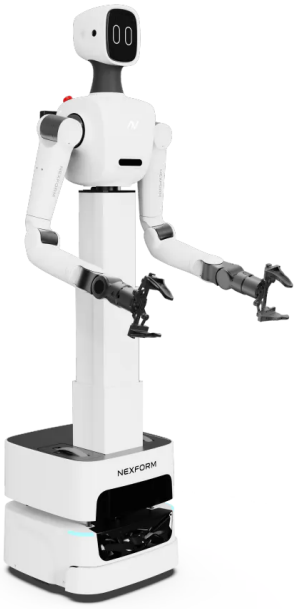
NEXFORM W1

Wheeled Dual-Arm Robot Product Manual



NEXFORM ROBOTICS

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Revision History

Version	Revision Date	Revision Notes
V0.2	2026-09-09	Rebuilt as the NEXFORM W1 Wheeled Dual-Arm Robot Product Manual based on the official W1 page data and component specifications (covering the PRO / ULTRA mobile models)

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1 Preface

The NEXFORM W1 wheeled dual-arm robot (PRO / ULTRA) is a **wheeled dual-arm embodied work robot** designed by NEXFORM ROBOTICS for commercial service, research and education, logistics and warehousing, and mobile manipulation scenarios. The complete machine consists of a liftable torso (dual arms plus neck/waist DoF) and an omnidirectional differential-drive mobile chassis, with 21 DoF in total across the body (excluding end effectors). It supports free switching between two modes, **fully autonomous operation** and **real-time teleoperation**, and can replace human labour in hotel service, commercial guidance, logistics sorting, supermarket handling, research and education, data collection, and home assistance scenarios to perform grasping, handling, inspection, and interaction tasks.

The product supports fixed / mobile dual-mode deployment: after removing the chassis it can serve as a fixed desktop dual-arm robot (repeat positioning accuracy ±0.1 mm); once the chassis is installed it becomes an autonomous mobile robot (SLAM mapping, autonomous navigation, and obstacle avoidance). This manual covers the **W1 wheeled models (PRO / ULTRA)** and describes their composition, technical specifications, safety rules, power supply and charging, and use and maintenance. Product parameters are subject to the official product page (nexform.tech/w1); the specific delivered configuration is subject to the sales list.

1.1 Hardware Composition

The standard delivery kit consists of the following (the final sales configuration list prevails):

No.	Item	Description
0	Complete robot body	W1 wheeled dual-arm robot (including dual arms, torso, mobile chassis, and lift column); aluminum alloy frame + plastic shell
1	Hot-swappable battery pack	2 units, tool-free 2-minute hot-swap battery replacement (ternary lithium battery, 24 Ah each)
2	Power adapter / charger	25.2V 10A; supports 100~240V AC input
3	Charging dock	Optional, used for autonomous recharging
4	End effector	Electric gripper / dexterous hand, optional (flange quick-change)

No.	Item	Description
5	Display / interaction components	Facial interaction screen (standard on ULTRA, standard on PRO), 6-element circular microphone array, dual speakers
6	Communication accessories	Wi-Fi 6 / Bluetooth 5.3 built in; 5G and GPS modules are standard on PRO / ULTRA

The main internal subsystems of the machine are as follows:

Layer	Composition
Execution layer (torso)	Two 7-DoF robotic arms, neck and waist DoF, end gripper/dexterous hand
Motion layer (chassis)	Differential-drive mobile chassis (two-wheel differential, 360° in-place rotation), lift column, battery compartment
Perception layer	Stereo wide-angle camera, fisheye cameras (ULTRA×3), waist RGB-D camera, wrist RGB-D cameras (ULTRA×2), LiDAR, low-obstacle detection, collision/magnetic sensors, ultrasonic sensors (ULTRA)
Computing layer	Base compute board (PRO 80 TOPS / ULTRA 128 TOPS, INT8), optional high-compute board, 24 GB memory, 64 GB storage (expandable)

1.2 Units

The physical quantity units used in this document and the SDK are defined as follows:

Parameter	Unit	Description
X / Y / Z	m	Position, metres
Dimensions (length×width×height)	mm	Overall machine dimensions, millimetres
Joint (joint angle)	rad / °	Joint angle, radians or degrees (expressed in ° in the joint motion range table)
v (movement/end-effector speed)	m/s	Metres per second
a (acceleration)	m/s^2	Metres per second squared
Angle (point-to-point accuracy, etc.)	°	Degrees
Mass / payload	kg	Kilograms

Parameter	Unit	Description
Torque	N·m	Newton-metres
Voltage / power consumption	V / W	Volts / watts
Current	A	Amperes
Battery energy	Ah	Ampere-hours
Compute	TOPS / TFLOPS	Trillions of integer / floating-point operations per second

1.3 Safety Instructions

1.3.1 Introduction

This section describes the safety principles and rules that must be followed when using the W1 robot and system. Before installation, use, or maintenance, users must carefully read this manual; content marked with warning symbols must be understood and strictly observed. Potential hazards such as pinching, collision, tipping, falling, and being crushed by motion exist during robot system operation; the mobile chassis has autonomous movement capability, so do not stand in its path of motion or place unrelated obstacles there. Users should fully understand the operational risks and use the SDK or control software only after training.

1.3.2 Safety Warning Symbols

Safety notices in other chapters of this manual use the following warning symbols, illustrated in Figure 1 (the actual labels supplied with the product prevail):

- **Danger**: used for hazardous situations that may result in death or serious equipment damage.
- **Warning**: used for hazardous situations that, if not avoided, could result in death or serious equipment damage.
- **High Temperature**: used for hazardous situations involving high-temperature parts (high-current charging/discharging components) where contact may cause burns or equipment damage.
- **Caution**: used for general warnings that, if not avoided, may result in personal injury or equipment damage.



图 1. Safety Warning Symbols

1.3.3 Safety Precautions

The following basic information must be understood and followed before starting the robot or robot system for the first time.

1.3.3.1 Robot Operation Precautions

1. The robot body and electrical equipment must be installed and fixed in accordance with the requirements of this manual; confirm that the complete machine (including the chassis) is placed stably and that there are no people or obstacles within its range of motion.
2. Before first use, have a professional perform a preliminary inspection of the robot and its protective systems, and test all safety functions (emergency stop, limits, collision protection, enable status, obstacle avoidance sensors).
3. Before use, check the integrity of the equipment and the tightness of installation, and confirm compliance with the local workplace safety regulations.
4. **The mobile chassis has autonomous movement capability.** Before powering on and enabling navigation / motion, confirm that there are no people, pets, or easily entangled obstacles in the chassis path; take care against pinching in narrow passages, doorways, and similar locations.
5. Ensure that the robotic arms have sufficient movement space and that there are no people, obstacles, or easily entangled cables within the range of motion; when both arms work simultaneously, take care against interference between the two arms and the torso.
6. **After disabling (`disable()`) or an abnormal power loss, the joints no longer actively output torque, and the robotic arms may fall under gravity.** Before performing related operations, return the arms to a safe pose first, or enter zero-gravity mode and support them by hand, to avoid death or equipment damage caused by falling from height.

7. If an accident or abnormality occurs during operation, press the hardware emergency stop / cut off the power immediately, then troubleshoot the problem.
8. Unauthorized personnel must not arbitrarily modify robot parameters through the SDK (joint limits, zero offsets, payload, mounting orientation, collision configuration, chassis speed limits, etc.); some parameter changes require a new safety assessment.

1.3.3.2 Personnel Safety

1. Users should carefully read the product manual and master safe, standard operating procedures and error handling methods.
2. While the equipment is running, even if the robotic arm or chassis appears to have stopped, it may be waiting for commands and should be regarded as still in motion.
3. Clearly define the range of motion of the robotic arms, end effectors (grippers, dexterous hands, etc.), and chassis, and set up a safe working area; do not stand directly in front of the robot or approach suddenly from the side during autonomous navigation.
4. Inspect the robot regularly to prevent loosening of fixing bolts, end-of-arm tools, chassis tyres, and battery latches.
5. When performing drag teaching (zero-gravity mode) or lifting/carrying the robot manually, be aware of its own weight and inertia, and avoid injury from sudden loss of force; after releasing both hands, the robotic arms should be able to hold their current pose — if they cannot, do not continue operating.

1.3.3.3 Battery and Charging Safety

1. Use the supplied / officially approved charger (25.2V 10A) and charging dock; never charge with a mismatched power adapter; the charger input is 100~240V AC.
2. Before charging or a battery swap, confirm that the cells and battery packs show no swelling, damage, or leakage; if any abnormality is found, stop charging immediately and contact after-sales service.
3. Hot-swap battery replacement must be performed while powered off (shut down or standby); hold the battery handle and pull it out along the guide rail, then insert it until the latch clicks into place, avoiding dropping, impacts, and short circuits.
4. Batteries are dangerous goods; do not disassemble, puncture, crush, immerse in water, or incinerate them; dispose of used batteries through professional recycling institutions in accordance with local regulations.
5. Keep the charging area ventilated and dry, away from flammable materials; if the battery temperature rises abnormally during charging, disconnect the input power immediately.

1.3.4 Responsibility and Regulations

The robot can form a complete system with other equipment; this document does not cover all design, installation, and operation content of a complete system. The safety of a complete system installation depends on the integration method; users must, in accordance with the laws, regulations, and safety standards of their country/region, carry out risk assessments for the design and installation of the complete system and take corresponding protective measures.

Starting to use this robot is deemed as having read, understood, and accepted all terms of this manual and the safety information. Users undertake responsibility for their own actions and consequences and use the robot only for legitimate purposes. We assume no liability for personal injury, accidents, property damage, etc. caused by violation of the usage requirements or by force majeure. Users must perform, but are not limited to, the following: risk assessment of the complete system, preparation of operating procedures, establishment of safety measures, confirmation that the system design and installation are correct, and no unauthorized modification of safety measures.

1.3.5 Risk Assessment

Risk assessment should consider all potential contacts between people and the robot under normal use and foreseeable misuse, for example:

- the hazard of using sharp end effectors or tool connectors;
- the hazard of handling toxic or otherwise hazardous substances;
- the hazard of fingers being pinched by arm joints, chassis wheels, or the lift column;
- the hazard of being struck by the robotic arm, or hit or run over by the mobile chassis;
- the hazard of the robot (including after the lift column is raised) tipping over or falling, or of end-of-arm tools not being securely fixed or loads exceeding limits and falling;
- the hazard of short circuits, overheating, or fire caused by electrical faults in the battery or charger;
- the hazard of cargo tipping over due to abnormal robot shutdown or emergency braking at low battery.

Users must determine through risk assessment whether the relevant hazards constitute unacceptable risks and take corresponding measures. This robot is not suitable for use by persons without professional guidance or without full civil capacity.

2 Technical Specifications

The technical specifications in this manual are based on the official W1 page (nexform.tech/w1) and provide a parameter comparison of the **two wheeled models, PRO / ULTRA**; items marked “optional” are determined by the actual configuration. The NANO / AIR desktop fixed models are outside the scope of this manual.

2.1 Mechanical Parameters

Parameter	PRO	ULTRA
Overall dimensions (column at lowest height)	429×460×1000 mm	429×460×1000 mm
Overall dimensions (column at highest height)	429×460×1650 mm	429×460×1650 mm
Material	Aluminum alloy frame + plastic shell	Aluminum alloy frame + plastic shell
Weight	62 kg	70 kg
Full-body DoF (excluding end effectors)	21	21
Neck DoF	3	3
Single-arm DoF	7	7
Waist DoF	1	1
Chassis DoF	2	2
Column DoF	1	1
Manual lift column	— (electric lift)	— (electric lift)

The full-body DoF is: neck 3 + two arms 7×2 + waist 1 + column 1 + chassis 2 = **21 DoF**. The column lift travel is approximately **1.0 m → 1.65 m** (the overall machine height changes with column extension).

2.2 Joint Motion Range

The joint motion ranges are identical for the PRO / ULTRA models. Arm joint ranges:

Joint	Range of Motion
J1	±160°
J2	±105°
J3	±162°
J4	-37° ~ 175°
J5	±162°
J6	±90°
J7	±90°

Torso / head-neck joints:

Joint	Range of Motion
Neck joint	P(-30°~45°), R(±30°), Y(±150°)
Waist joint	±175°
Joint output bearing	Crossed roller bearing

Joint	Range of Motion
Cable routing	Internal routing

2.3 Performance Parameters

Parameter	PRO / ULTRA
Single-arm payload	3 ~ 6 kg
Maximum shoulder torque	97 N·m
Repeat positioning accuracy	±0.1 mm
Maximum arm end-effector speed	3 m/s
Maximum arm end-effector acceleration	0.8 m/s²
Compatible end effector types	Gripper / dexterous hand

2.4 Chassis Mobility Parameters

Parameter	PRO	ULTRA
Chassis travel speed	Max 1.2 m/s (default 0.7 m/s)	Max 1.2 m/s (default 0.7 m/s)
Chassis drive type	Two-wheel differential, supports 360° in-place rotation	Two-wheel differential, supports 360° in-place rotation
Sensors	LiDAR ×1 + depth camera ×1 + low-obstacle detection ×1 + physical magnetic sensor ×2 + physical collision sensor ×2	LiDAR ×1 + depth camera ×2 + low-obstacle detection ×1 + physical magnetic sensor ×2 + physical collision sensor ×2 + ultrasonic sensor ×2
Mapping performance	300 m × 300 m (5 cm resolution)	500 m × 500 m (5 cm resolution) / 350 m × 350 m (1.5 cm resolution)
Point-to-point accuracy (average)	±5 cm (5 cm resolution)	±2 cm (1.5 cm resolution)
Point-to-point accuracy (maximum)	±8 cm (5 cm resolution)	±4 cm (1.5 cm resolution)
Minimum point-to-point angle	±3.0°	±1.0°
Multi-robot obstacle avoidance	Optional	Supported

The chassis has SLAM autonomous mapping / navigation capability and supports 360° all-round perception and multi-sensor fusion obstacle avoidance; it can pass flexibly through narrow passages (minimum about 55 cm), low obstacles, doorways, and similar scenarios.

2.5 Perception and Interaction

Perception system

Parameter	PRO	ULTRA
Stereo wide-angle camera	1	1
Fisheye cameras	Optional	3
Head RGB-D camera	Optional	Optional
Waist RGB-D camera	1	1
Wrist RGB-D cameras	Optional	2

Interaction modules

Parameter	PRO / ULTRA
Microphone array	6-element circular array
Speaker	Dual speakers × 5 W
Facial interaction screen	Yes
Light strip	Yes

Communication and positioning

Parameter	PRO / ULTRA
Wi-Fi 6 / Bluetooth 5.3	Yes
5G	Yes
GPS	Yes

2.6 Power and Energy

Parameter	PRO / ULTRA
Number of batteries	3 (chassis battery ×1 + hot-swap batteries ×2)
Battery energy	Chassis 18.2 Ah + hot-swap 24 Ah × 2
Battery type	Ternary lithium battery
Endurance	About 7 hours
Energy replenishment methods	Manual charging / autonomous recharging / autonomous battery swap
Charging time	About 2 hours
Battery hot swap	Yes
Battery swap support	Yes
Charger	25.2 V 10 A

Parameter	PRO / ULTRA
Charging input voltage/current	100~240 V / 8 A
Charging output voltage/current	54.6 V / 8 A
Power supply mode	Battery powered
Battery output voltage	46.8 V
Battery maximum power	468 W
Power supply interface	24 V 3 A

The robot supports three energy replenishment methods; see the “Power Supply and Charging” chapter for details.

2.7 Computing Platform

Parameter	PRO	ULTRA
Base compute board	80 TOPS (INT8), 6-core Cortex-A78 @1.5 GHz	128 TOPS (INT8), 6-core Cortex-A78 @2.0 GHz
High-compute board / secondary development board	Optional	NVIDIA Jetson Thor T5000 (2070 TFLOPS FP4)
Memory	24 GB	24 GB
Storage	64 GB (expandable)	64 GB (expandable)
External interfaces	USB 3.0 (optional expansion dock)	USB 3.0 (optional expansion dock)

2.8 Software and Development

Parameter	PRO / ULTRA
Data collection-training-inference development platform	Supported (full data collection-training-inference workflow)
Robot management platform	Supported
Dual-arm teleoperation modes	Isomorphic teleoperation / VR teleoperation / software teleoperation
Dual-arm safety boundary protection	Supported
Secondary development SDK	Supported
OTA upgrade	Supported
Simulation platforms	Isaac Sim, Gazebo, MuJoCo, PyBullet
Software compatibility	Python / C++ / ROS1 / ROS2 / MoveIt
Supported models	VLAD, ACT, π_0 (Pi0), GR00T

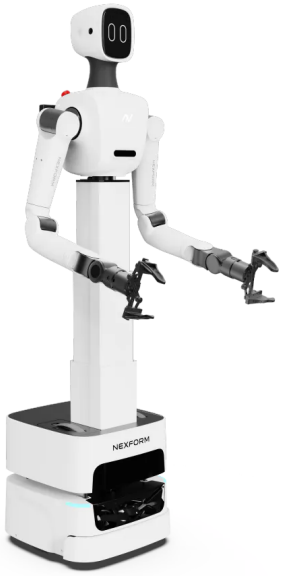


图 2. W1 complete machine (official website)

2.9 Illustrations

The following figures show the overall view of the W1 and the annotated functional overview of the machine (images from the official website).

3 Power Supply and Charging

3.1 Electrical Safety

1. Do not plug or unplug the battery, charging cables, or actuator cables while powered; shut down and disconnect power before installation, battery swap, and maintenance work.
2. Observe water and moisture protection; if water enters, shut down immediately, disconnect power, and contact after-sales service.
3. Charging must use the supplied / officially approved charger (25.2V 10A, 100~240V AC input); the input/output polarity, voltage, and current must meet the specification requirements.
4. Do not reverse the positive and negative wiring, otherwise the battery and the machine may be damaged; confirm that the robot body and the power ground are reliably connected.



图 3. W1 complete machine functional overview annotation (official website)

3.2 Power Architecture

The W1 adopts a “chassis battery + dual hot-swap batteries” power architecture: the chassis battery supplies the mobile chassis and actuators; the two hot-swap batteries support tool-free replacement, and the whole machine does not need to be shut down during a battery swap. The robot supports three energy replenishment methods, covering fixed workstations, long-duration tasks, and unattended scenarios:

- **Manual charging (wired)**: plug in at a fixed workstation (charger 25.2V 10A) for operation / charging with full compute power;
- **Autonomous recharging**: when the battery level drops below 20%, automatically navigate to the charging dock to charge;
- **Autonomous battery swap**: hot-swap battery replacement without shutdown; the battery pack can be replaced in about 2 minutes (performed by a battery-swap robot / personnel).

3.3 Charging Specifications

Item	Parameter
Charger output	25.2 V 10 A
Charging input	100~240 V AC, 8 A
Charging output (battery side)	54.6 V / 8 A
Charging time (0→100%)	About 2 hours

Item	Parameter
Battery output voltage	46.8 V
Battery maximum power	468 W

4 Usage Guide

4.1 Introduction

Before powering on the W1 wheeled dual-arm robot, perform the checks in Section 1.4, Safety Instructions. Standard usage procedure:

1. Unpack and inspect, confirming that accessories are complete and undamaged;
2. Install / check the hot-swap batteries and confirm sufficient charge (or connect the charger / dock at the charging station);
3. Determine the work site and confirm that the chassis range of motion and the arm movement space are free of obstacles;
4. Power on, and observe that the self-check and indicator light status are normal;
5. Complete preparation according to the deployment form (fixed desktop / mobile chassis);
6. Access via software (autonomous operation / teleoperation) to perform tasks.

4.2 Appearance and Component Overview

The overall appearance and component annotations of the machine are shown in Figure 2 and Figure 3; the main components are described in the following table:

Component	Description
Mobile chassis	Differential drive, 360° in-place rotation; integrates LiDAR, depth cameras, low-obstacle detection, magnetic / collision / ultrasonic sensors
Lift column	Electric lift, travel approximately 1.0 m → 1.65 m, enabling the dual arms to reach targets from the floor to overhead height
Torso	Includes waist DoF and dual-arm mounting interfaces; arms have 7 DoF, internal routing
End flange	For mounting end-of-arm tools such as grippers / dexterous hands
Head	Stereo wide-angle camera, fisheye cameras (ULTRA), microphone array, facial interaction screen
Battery compartment	Chassis battery + dual hot-swap battery packs
Interaction screen / light strip	Status and interaction display, operating light effects

4.3 Unpacking and Installation

4.3.1 Unpacking Inspection

1. Check whether the outer packaging is damaged; after unpacking, check against the packing list to confirm that the complete machine, battery packs, charger, and accessories are all present;
2. Check that the machine exterior has no impacts, deformation, or damage, and that cables are not damaged or exposed;
3. Check that the battery packs are intact in appearance, insert the hot-swap batteries into the battery compartment along the guide rails, and confirm that the latches are in place;
4. Properly keep the packaging materials for return repair / transport.

4.3.2 Deployment Forms

The W1 supports fixed / mobile dual-mode deployment:

- **Form A (fixed desktop)**: with the mobile chassis removed / not installed, fix the torso to a desktop or machine bench so it becomes a precision assembly station, used for high-precision (±0.1 mm) assembly, sorting, and other fixed-workstation operations;
- **Form B (mobile autonomous)**: install the differential-drive mobile chassis and perform SLAM mapping (maximum 500 m × 500 m) to enable autonomous navigation, delivery, inspection, and other mobile operations.

4.3.3 Power-On and Power-Off

1. Confirm that the batteries are installed and sufficiently charged, and that there are no people / obstacles within the robot's range of motion;
2. Turn on the main power / press the power button, wait for the system self-check to complete, and the indicator light enters the ready state;
3. Before use, confirm that the emergency stop can be triggered normally and that the collision and obstacle avoidance protection functions work properly;
4. After use, stop the task / disconnect the software connection first, then power off; if not used for a long time, remove the hot-swap batteries.

4.3.4 Charging and Battery Swap Operations

- Manual charging: connect the charger's DC output to the charging port (or the chassis charging interface) and the AC side to a 100~240V power supply; a full charge takes about 2 hours; the indicator stays on during charging and changes to steady / a different colour when full (subject to the actual indicator).
- Autonomous recharging: after enabling the recharging function on the management platform, the robot automatically navigates to the charging dock when the battery level drops below 20%; keep the recharging route clear and the charging contacts clean.
- Hot-swap battery replacement: with the robot powered off (or in standby), press / flip open the battery latch, hold the handle to pull out the battery, then insert a fully charged battery until the latch is in place. The swap process takes about 2 minutes and requires no disassembly of the machine.

4.4 Software Access and Teleoperation

The robot software supports multiple access methods (see the corresponding development manual / software manual for details):

1. Connect to the machine from a browser / desktop via the robot management platform for device management, task dispatch, and status monitoring;
2. Perform secondary development via the SDK (Python / C++) or ROS1 / ROS2 / MoveIt, and access arm, chassis, and camera data;
3. Teleoperation supports three modes: **isomorphic teleoperation, VR teleoperation, and software teleoperation**; dual-arm safety boundary protection is active during teleoperation to avoid collisions with people and obstacles;
4. For embodied intelligence development, a built-in data collection-training-inference platform supports closed-loop training for imitation learning (ACT / VLAD / π_0 / GR00T, etc.) and import into Isaac Sim / Gazebo / MuJoCo simulation.

4.5 Maintenance

Item	Cycle / Frequency	Content
Exterior cleaning	Weekly	Clean the shell, screen, and light strip with a dry soft cloth; avoid direct liquid spraying
Mechanical inspection	Monthly	Check fixing bolts, end-of-arm tools, battery latches, chassis tyres, and column guide rails

Item	Cycle / Frequency	Content
Sensors	Monthly	Clean the surfaces of LiDAR, camera lenses, and obstacle avoidance sensors; check that protective covers are unobstructed
Battery	Every battery swap	Check that the battery appearance shows no swelling / damage / leakage, and that contacts are clean and free of oxidation
Charging contacts	Weekly	Clean the contacts of the charging dock and charging port; remove dust and grease
System	As needed	Update the system and firmware via OTA, and back up task and map data

4.6 FAQ and Technical Support

4.6.1 Q&A

Q: What control / operating modes does the robot have? A: The W1 supports free switching between fully autonomous operation (autonomous navigation + dual-arm tasks) and real-time teleoperation (isomorphic / VR / software teleoperation), and the chassis can be removed for use as a fixed desktop model. Before moving, confirm that each joint is enabled and that it is not in the emergency stop state.

Q: What is the difference between joint space and Cartesian space? A: Joint space directly controls each joint angle; it is simple and direct, less prone to singularities, and suitable for point-to-point motion. Cartesian space controls the end-effector pose and is suitable for tasks requiring precise trajectories such as straight lines / arcs; planning is more complex and may encounter singularities (see the development manual for details).

Q: The robot cannot move after power-on / reset? A: Please check in order: whether the software and robot are networked; whether it is in the emergency stop (stop) or fault (fault) state; whether the joints / chassis are enabled; whether the battery is sufficiently charged. Following the safety procedure, execute `clear_stop()`, `clear_faults()`, and `enable()` in order before moving.

Q: How do I perform drag teaching safely? A: After entering zero-gravity mode (`zero_gravity()`), slowly drag the robotic arms by hand to the target pose; do not drag beyond the joint soft limits; after stopping the drag, confirm that the arms can hold their pose; after teaching is complete, exit zero-gravity mode and have the arms hold / return to a safe pose.

Q: Why do the robotic arms drop after `disable()`? A: After disabling, the joints no longer output torque, and the arms will fall under gravity. Operate when the arms are supported and no one is nearby, or return them to a safe pose first.

Q: What should I do if the robot has insufficient endurance or cannot localize for a long time? A: Check the battery level and charging / battery swap; confirm that mapping is complete and that the current environment features match the map, and re-map if necessary; avoid degraded scenarios such as highly reflective glass and empty corridors that cause localization drift; the ULTRA model can use a higher-resolution map (1.5 cm) to improve localization accuracy.

Q: What should I do if I cannot connect to the robot? A: Confirm that the robot and the host computer are on the same network (Wi-Fi 6 or Ethernet), and verify the IP and port configuration; restart the robot and the control service if necessary. If the problem persists, contact technical support.

Q: What are the technical support channels? A: The NEXFORM ROBOTICS technical team; the official website nexform.tech (Service and Support); the GitHub organization github.com/nexform-tech (Issues).

End of document If you have any questions or need additional information, please contact the NEXFORM ROBOTICS technical team.